A PRODUCT OF PAIRS IS HENSELIAN EXACTLY WHEN EACH FACTOR IS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The henselian property is preserved by integral ring maps. Lemma \ref{lemma-integral-over-henselian-pair}, cited at proof lines 2817-2819. Gives the direction from the product to the factors, the projections being integral.
- Elements of one plus the ideal of a henselian pair are units, and an ideal whose elements of that form are units lies in the Jacobson radical. Algebra, Lemma \ref{algebra-lemma-contained-in-radical} and Definition \ref{definition-henselian-pair}, cited at proof lines 2823-2826 and again at 2828. Show that the product ideal is contained in the Jacobson radical of the product ring.
- Definition of a henselian pair: liftability of factorisations of monic polynomials modulo the ideal. Definition \ref{definition-henselian-pair}, cited at proof lines 2839-2840. Is the property verified componentwise for the product pair.

Source reference. Original lemma 0ATD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13611–13611. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 09XK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12533–12533. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0AME: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13402–13402. Navigation pointer to a statement the

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Source reference. Original lemma 09XE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12527–12527. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let J be a set and let $\{(A_j, I_j)\}_{j\in J}$ be a collection of pairs. Then $(\prod_{j\in J} A_j, \prod_{j\in J} I_j)$ is Henselian if and only if so is each (A_j, I_j).*

Proof. For every $j \in J$, the projection $\prod_{j\in J} A_j \to A_j$ is an integral ring map, so Lemma 09XK proves that each (A_j, I_j) is Henselian if $(\prod_{j\in J} A_j, \prod_{j\in J} I_j)$ is Henselian.

Conversely, suppose that each (A_j, I_j) is a Henselian pair. Then every $1 + x$ with $x \in \prod_{j\in J} I_j$ is a unit in $\prod_{j\in J} A_j$ because it is so componentwise by Algebra, Lemma 0AME and Definition 09XE. Thus, by Algebra, Lemma 0AME again, $\prod_{j\in J} I_j$ is contained in the Jacobson radical of $\prod_{j\in J} A_j$. Continuing to work componentwise, it likewise follows that for every monic $f \in (\prod_{j\in J} A_j)[T]$ and every factorization $\overline{f} = g_0 h_0$ with monic $g_0, h_0 \in (\prod_{j\in J} A_j/\prod_{j\in J} I_j)[T] = (\prod_{j\in J} A_j/I_j)[T]$ that generate the unit ideal in $(\prod_{j\in J} A_j/\prod_{j\in J} I_j)[T]$, there exists a factorization $f = gh$ in $(\prod_{j\in J} A_j)[T]$ with g, h monic and reducing to g_0, h_0. In conclusion, according to Definition 09XE $(\prod_{j\in J} A_j, \prod_{j\in J} I_j)$ is a Henselian pair. □